Hydroelectric Energy

Chapter 6

Like wind energy, water energy has been harnessed for thousands of years. It has always been used to produce mechanical energy, the energy of motion. But only recently in human history has moving water been used to produce electrical energy. **Hydroelectric power** is an important renewable energy resource.

Old water mills converted the energy in moving water into mechanical energy. A modern hydroelectric power turbine does the same thing.

Big Question

How do people use moving water as a source of energy?

Vocabulary

hydroelectric power, n. electricity generated by the energy in moving water, usually through a dam

Hydroelectric power is more predictable than wind power because water can be contained, or controlled. A hydroelectric power plant involves a dam that creates a large reservoir. The reservoir provides a steady supply of water for the turbines.



Hydroelectric power involves first positioning water above turbines. Gravity then pulls the water through the turbines.

Things to Consider About Hydroelectric Power

A hydroelectric dam can be set up where there is a lot of water uphill from an area where it can flow down safely. It is most practical to set up a dam where flowing water can steadily refill a reservoir. This is why hydroelectric dams are usually built on large rivers.

Engineers must consider the source of the river water. The water cycle brings water onto land in the form of precipitation that falls when clouds cool and condense. When rain falls, it flows downhill and builds up in streams. The same occurs when snow and ice melt. But some of the flow from melting ice can be from precipitation that fell long ago and accumulated over a very long time. A glacier that is slowly melting and feeding water into a river



This hydroelectric dam in Austria relies on seasonal snowfall that becomes meltwater, which fills the reservoirs. Without winter precipitation, the dam would be less productive.

may eventually melt away completely. Then the river could shrink to a trickle or vanish completely. A more reliable river over the long term might be one that receives water from rainfall and snow that melts each spring and summer. Either way, engineers must consider data about the weather and climate of a given area.

Hydroelectric power depends on large amounts of water spinning turbines to generate enough electricity to make the project **cost effective**. Blocking a river so that a large reservoir will form behind the dam might be necessary to make the project cost effective. But this takes a lot of time, money, and careful planning.

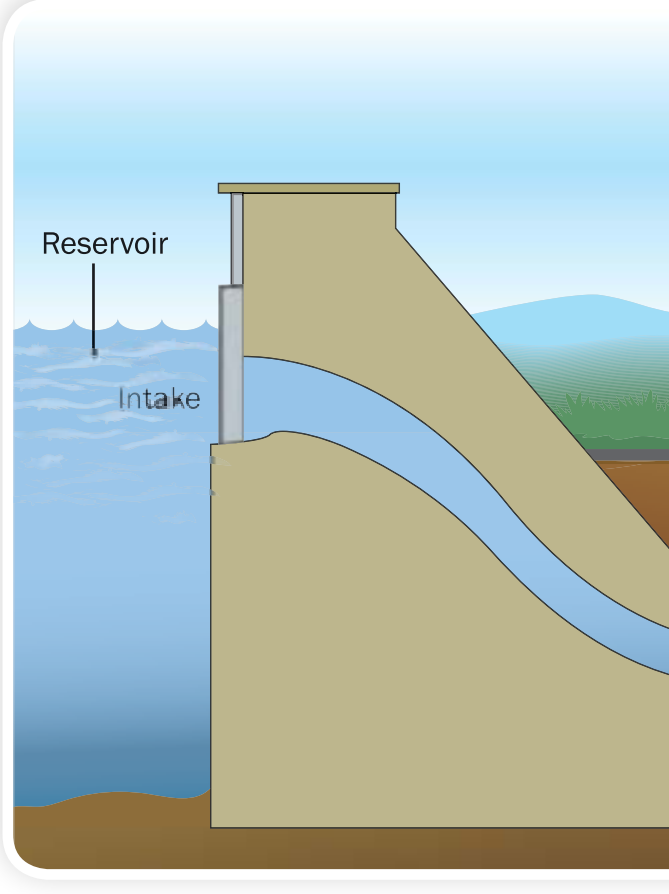
Vocabulary

cost–effective, adj. describing the balance of factors when benefits outweigh costs



Environmental Costs and Safety Hazards

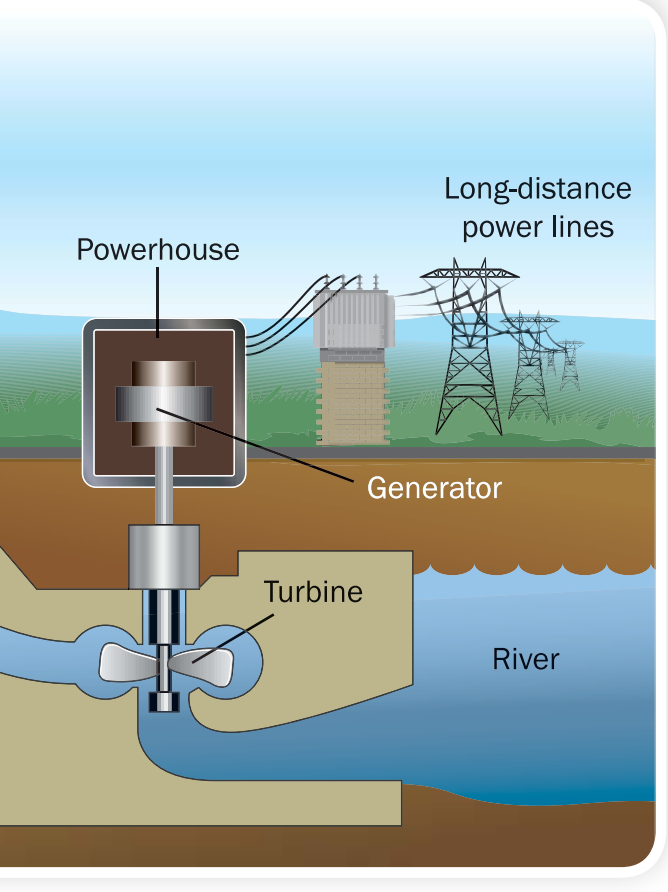
While hydroelectric power is very clean, it is not without costs to the environment. Blocking a river to produce a large reservoir floods the area behind the dam. This destroys some habitats. Dams can also interfere with fish such as salmon. Salmon are born in rivers and migrate to the sea. Then they return to the streams where they hatched to reproduce. A dam also alters the river downstream. The flow of the river may be slowed and its depth reduced. Organisms that once had a fast-moving, wide, deep river might have something slower and narrower to swim through than before. Or the river may change dramatically from one week to the next as the dam controls the flow of water to meet the region's electrical needs.



Hydroelectric dams generate electricity. Water moves from the reservoir into the intake. It spins a turbine, which is connected to a generator. The generator then generates electricity, which is sent through power lines to power homes and businesses.

Fish ladders can be built to help migrating salmon move up and down the dammed parts of the river. There are also newer dam and turbine designs that are "fish friendly." These allow fish to pass through with less interference. However, a dam's overall disturbance to river environments is hard to completely overcome.

Dams are also vulnerable to accidents. If a river or reservoir receives an unusually large amount of water from high rainfall, the pressure on its concrete and steel walls may be too great. This occurred in Laos in 2018, when a multibillion-dollar dam that was still under construction collapsed after heavy rains. A large amount of water suddenly spilled through the dam and flooded the landscape.



Hydroelectric Power and Tides

Water moved downhill by gravity is one renewable resource that can be harnessed. Water moved by tides is another. Tides are caused by gravity, too. But instead of pulling water downhill, tides pull ocean water across Earth's surface. Tides result from forces of gravity between the moon and the sun and Earth's water.

To harness tidal power, turbines can be placed in areas where tides produce a strong current. A shallow span of ocean is often the best place because tides tend to move swiftly there. A dam-like structure can also be installed in an area with strong tides. As the tide comes in, the turbines spin one way. When the tide goes out hours later, the turbines spin the other way, producing electricity.

Hydroelectric power that uses tidal movement is practical only in some areas. Because tidal movement changes speed and can slow to a standstill, an electric grid is unlikely to rely on tidal hydroelectric power alone.



Narrowing the area through which tides can flow makes transforming hydroelectric power into electricity more efficient.